

Study / Baseline Vector	Task Focus	Text Register	Embedding Strategy	Classification Paradigm	Core Limitation vs. MediTriapaA
Khan et al. (2021)	General Symptom Extraction	Pure Downstream Hindi	Static Embeddings (FastText)	CNN + BiLSTM	Falls completely on Romanized code-switching (Hinglish); cannot handle structural alternations.
Lai et al. (2022)	Social Media Sentiment (Hinglish)	Twitter / Reddit informal text	Word2Vec + Character-level CNN	SVM / Random Forest	Lacks clinical context; character-CNNs fail on phonetic mutations of anatomical terms (e.g., seena vs chine).
Singh & Gupta (2023)	Multi-lingual Medical QA	Parallel Corpora (En / Hi translations)	mBERT (Multilingual BERT)	Fine-tuned Transformer Heads	Assumes clean language boundaries; cross-lingual transfer collapses over conversational fragments (Type C).
Sharma et al. (2024)	Code-Mixed Health Dialogue	High-Formality Interpersonal Hinglish	XLNet+RoBERTa	Sequence-to-Sequence (T5)	Naive LLM generation resulted in homogeneous text; no programmatic phonetic perturbation or leakage protection.
MediTriapaA (Ours, 2026)	Dual-Task Severity & Specialist Triage	Trilingual + Programmatically Perturbed (Types A, B, C)	Contextual Suiword Trans-Encoder (XLNet / BanglaBERT-derived variants)	Decoupled Multi-Task Classification Heads with Deterministic Pre-processing Safety Gate	None. Resolves structural homogeneity via Moseken frameworks and protects evaluation metrics via seed-level split validation.